

TECHSTACK.md

SynChain Technology Stack

Overview

SynChain is designed as a scalable, AI-native supply chain intelligence platform. The technology stack prioritizes developer productivity, performance, maintainability, and future scalability.

Frontend

Framework

- Next.js 15
- React 19
- TypeScript

Styling

- Tailwind CSS
- Shadcn UI

Animations

- Framer Motion

Data Visualization

- Recharts
-

Backend

API Framework

- FastAPI

Language

- Python 3.12+

API Documentation

- OpenAPI

- Swagger UI
-

Database

Current Development

- SQLite

Production Database

- PostgreSQL

Reason:

- ACID compliance
 - High performance
 - Enterprise reliability
 - Excellent analytics support
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ORM

SQLAlchemy

Responsibilities:

- Database abstraction
 - Query generation
 - Schema management
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AI Layer

Agent Framework

- LangGraph

Responsibilities:

- Agent orchestration
- Workflow management
- State handling

LLM Providers

Supported:

- OpenAI
- Claude
- Gemini

Provider selection should remain configurable.

External Intelligence Layer

Search

- Tavily

Future Sources

- News APIs
 - Industry Reports
 - Market Intelligence Feeds
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Caching

Redis

Responsibilities:

- Agent memory
 - Temporary state storage
 - Result caching
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Background Processing

Celery

Responsibilities:

- Scheduled jobs
 - News ingestion
 - Agent execution
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Deployment

Frontend

- Vercel

Backend

- Railway

Database

- Supabase PostgreSQL
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Containerization

Docker

Used for:

- Development consistency
 - Production deployment
 - CI/CD pipelines
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Monitoring (Future)

- Grafana
 - Prometheus
 - Sentry
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Security

- JWT Authentication
 - HTTPS
 - API Rate Limiting
 - Role-Based Access Control
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Design Principle

Every technology choice must satisfy:

1. Scalability
2. Reliability
3. AI compatibility
4. Developer productivity
5. Low operational complexity